

THE LIMITS OF ABELIAN RECONSTRUCTION: TWIN CURVES EXIST, GENUS ONE SUFFICES, AND THE OBSTRUCTION IS A TORSOR COORDINATE

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ABSTRACT. Paper XIX of the series asked whether there are non-isomorphic curves over $\mathbb{F}_q$ with the same zeta function *and* isomorphic Jacobian groups, which would locate the resolution limit of the abelian Bost–Connes systems. The answer is yes, the construction is far simpler than the genus-two machinery proposed, and the reason is complex multiplication.

For an ordinary elliptic curve $E/\mathbb{F}_q$ with $\text{End}(E) = \mathcal{O}$, Deuring’s theory gives $E(\mathbb{F}_{q^n}) \cong \mathcal{O}/(\phi^n - 1)$ as an $\mathcal{O}$ -module for every n , an isomorphism depending only on $\mathcal{O}$; and $\text{Pic}(\mathcal{O})$ acts simply transitively on the set of curves in the isogeny class with that endomorphism ring, of which there are exactly $h(\mathcal{O})$. Hence whenever $h(\mathcal{O}) > 1$ one obtains $h(\mathcal{O})$ curves with distinct j -invariants sharing *every* Galois-module invariant at every level. Genus one already suffices; Howe’s and Lauter’s genus-two constructions are not needed. Enumerating all curves over $\mathbb{F}_p$ for $11 \leq p \leq 29$ we find 69 isogeny classes containing several j -invariants with a common group structure, and the counts match the class numbers: $h(-88) = h(-100) = h(-48) = h(-32) = h(-60) = 2$ against two j -invariants in each case.

This is stronger than the question asked. Twin curves agree not only in Z_C and in $\text{Jac}(\mathbb{F}_q)$, but in the whole tower $\{E(\mathbb{F}_{q^n})\}_n$ with its trace maps — which matters because Tier 2 of Paper XIX recovers more than the abstract group: the compactified Picard group with its translation structure, through the deep stratum, and with it the map $P \mapsto [P] \in \text{Pic}(C)$ on all closed points. We prove that this map does not separate twins either — under the module identification the class of a degree- n point is multiplication by $(\phi^n - 1)/(\phi - 1)$, again $\mathcal{O}$ -determined — and confirm it numerically: for two supersingular classes at $p = 11, 17$ and for each group structure at $p = 13$, the degree-two point-to-Picard multiset profiles coincide across distinct j -invariants.

We also correct the proposed explanation of the barrier. The Weil pairing is not out of abelian reach for the reason given: over $\mathbb{F}_q$ the Galois group acting on $\text{Jac}[n]$ is $\text{Gal}(\mathbb{F}_q/\mathbb{F}_q) = \hat{\mathbb{Z}}$, which is abelian, so “the pairing cannot be encoded in an abelianized Galois group” is not the obstruction. The obstruction is a torsor coordinate: $\text{Pic}(\mathcal{O})$ acts simply transitively on the twins and acts *trivially* on every module invariant, so what is missing is the position in a $\text{Pic}(\mathcal{O})$ -torsor, and no invariant of the modules can see it.

1. TWIN CURVES, AND WHY GENUS ONE SUFFICES

The framework is that of [6] transported to function fields as in Papers XVII–XIX of the series; Papers II, IX, X, XIV and XVI–XIX, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place.

Theorem 1.1 (Existence of twins). *Let $E/\mathbb{F}_q$ be an ordinary elliptic curve with Frobenius ϕ and $\text{End}(E) = \mathcal{O}$, an order in $\mathbb{Q}(\phi)$ with $\mathbb{Z}[\phi] \subseteq \mathcal{O} \subseteq \mathcal{O}_{\max}$. Then:*

- (1) $E(\mathbb{F}_{q^n}) \cong \mathcal{O}/(\phi^n - 1)$ as an $\mathcal{O}$ -module, for every $n \geq 1$, and this depends only on $\mathcal{O}$ and ϕ , not on E ;
- (2) $\text{Pic}(\mathcal{O})$ acts simply transitively on the set of $\mathbb{F}_q$ -isomorphism classes of curves in the isogeny class with endomorphism ring $\mathcal{O}$ — equivalently, since the quadratic twist has trace $-t \neq t$ for ordinary E , on their j -invariants; there are exactly $h(\mathcal{O})$ of them.

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XX-twin-curves/](#).

Consequently, whenever $h(\mathcal{O}) > 1$ there exist $C_1 \not\cong C_2$ over $\mathbb{F}_q$ with $Z_{C_1} = Z_{C_2}$, with $\text{Jac}(C_1)(\mathbb{F}_q) \cong \text{Jac}(C_2)(\mathbb{F}_q)$, and indeed with $C_1(\mathbb{F}_{q^n})$ and $C_2(\mathbb{F}_{q^n})$ isomorphic as $\mathcal{O}$ -modules for every n .

Proof. (1) is Deuring's theory [1] in the module-theoretic form proved by Lenstra [4]: for ordinary E with $\text{End}(E) = \mathcal{O}$ one has $E(\mathbb{F}_{q^n}) = \ker(\phi^n - 1) \cong \mathcal{O}/(\phi^n - 1)$ as $\mathcal{O}$ -modules. (2) is the theory of complex multiplication in characteristic p , in Waterhouse's kernel-ideal form [5]: the set of curves with a given ordinary endomorphism ring is a principal homogeneous space under the class group. Distinct classes give distinct j -invariants, these being the roots of the Hilbert class polynomial of $\mathcal{O}$ reduced mod p . $\square$

Remark 1.2 (The proposal over-engineers). The construction requested — genus-two hyperelliptic curves with equal Igusa invariants data, following Howe [2] and Lauter [3] — is unnecessary. Genus one produces twins as soon as $h(\mathcal{O}) > 1$, which is the generic situation, and produces them with a stronger property than asked for: agreement of the entire Galois-module tower, not merely of Z_C and $\text{Jac}(\mathbb{F}_q)$.

Example 1.3 (Enumeration). Enumerating all $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$ for $11 \leq p \leq 29$ and grouping by (trace, group structure) yields 69 classes containing more than one j -invariant. Samples, with the class number of the relevant order:

p	t	$\text{Jac}(\mathbb{F}_p)$	j -invariants	$h(\mathcal{O})$
23	-2	$\mathbb{Z}/26$	1, 17	$h(-88) = 2$
13	2	$\mathbb{Z}/12$	4, 6	$h(-48) = 2$
17	-6	$\mathbb{Z}/24$	1, 6	$h(-32) = 2$
19	-4	$\mathbb{Z}/24$	9, 17	$h(-60) = 2$
29	-4	$\mathbb{Z}/34$	3, 7, 17	$h(-100) = 2$ (two orders)
23	-6	$\mathbb{Z}/30$	14, 18, 20, 22	$h(-56) = 4$

The count of j -invariants at a fixed group structure equals $h(\mathcal{O})$ for the corresponding endomorphism ring, as Theorem 1.1(2) requires.

2. INDISTINGUISHABILITY AT BOTH TIERS

Theorem 2.1 (Twins are invisible to the abelian system). *Let C_1, C_2 be twins in the sense of Theorem 1.1, with a common endomorphism ring $\mathcal{O}$. Then their Bost–Connes systems agree at both tiers of Theorem 4.1 of Paper XIX:*

- (1) Tier 1. *The partition functions coincide, $Z_{C_1} = Z_{C_2}$.*
- (2) Tier 2. *$\text{Jac}(C_1)(\mathbb{F}_q) \cong \text{Jac}(C_2)(\mathbb{F}_q)$; more, the whole tower $\{C_i(\mathbb{F}_{q^n})\}_n$ with its norm maps is isomorphic, so every invariant of the Galois-module data agrees; and more still, the full Tier-2 datum — the compactified Picard group together with the degree-labelled multiset of point classes $\{[P] : \deg P = n\}$ — corresponds under the module identification, for every n .*

Proof. Both follow from Theorem 1.1(1): the tower and its maps are $\{\mathcal{O}/(\phi^n - 1)\}$ with the natural maps, depending only on $(\mathcal{O}, \phi)$. For the last clause of (2): a closed point of degree n is a Frobenius orbit $\{Q, \phi Q, \dots, \phi^{n-1}Q\}$ of a point $Q \in C(\mathbb{F}_{q^n})$ of exact degree n , and, choosing the rational base point, its class in $\text{Pic}^0 = E(\mathbb{F}_q)$ is $Q \oplus \phi Q \oplus \dots \oplus \phi^{n-1}Q = \nu_n \cdot Q$ with $\nu_n = (\phi^n - 1)/(\phi - 1) \in \mathcal{O}$, computed in the $\mathcal{O}$ -module $\mathcal{O}/(\phi^n - 1)$, with $E(\mathbb{F}_q) = \ker(\phi - 1) = \nu_n \cdot \mathcal{O}/(\phi^n - 1)$. The degree- n multiset of point classes is therefore the image multiset of ν_n on the exact-degree- n elements of $\mathcal{O}/(\phi^n - 1)$ — a formula in $(\mathcal{O}, \phi)$ alone. Since the Tier-2 datum recovered in Theorem 3.1 of Paper XIX is exactly the compactified Picard group with these degree-labelled translations, the twins' data are isomorphic as marked groups. $\square$

Remark 2.2 (Tier 2 recovers more than the abstract group). It is worth being careful here, because the question as posed assumed Tier 2 sees only $\text{Jac}(\mathbb{F}_q)$. It sees more. By Theorem 2.1 of Paper XIX, $G_S/\overline{U_S} \cong \widehat{\text{Pic}(C)}$, and the deep-stratum translation structure recovered in the proof of Theorem 3.1 of Paper XIX — the generator $P \in J_S$ acting on the Picard torsor by σ_P — gives the map

$$P \mapsto [P] \in \text{Pic}(C)$$

on all closed points, together with their degrees. That is substantially finer than the group $\text{Jac}(\mathbb{F}_q)$, and it is what has to be checked. By Theorem 2.1(2) it is also $\mathcal{O}$ -determined, so twins are not separated by it either.

Remark 2.3 (Numerical check of the point-to-Picard map). For an elliptic curve with base point O , a closed point of degree two is a Galois orbit $\{Q, \phi Q\}$ in $E(\mathbb{F}_{q^2})$ and its class in $\text{Pic}^0 = E(\mathbb{F}_q)$ is $\text{Tr}(Q) = Q \oplus \phi Q$. The last clause of Theorem 2.1(2) predicts equal profiles across twins; we confirm it by direct computation, for the ordinary twins of Example 1.3 at $p = 13$ and also for two supersingular same-zeta-same-group classes (outside the ordinary hypothesis of Theorem 1.1, so not covered by the proof):

p	t	j	degree-two profile
11	0	0	$(6^6, 5^6)$
11	0	1	$(6^6, 5^6)$
17	0	0	$(9^9, 8^9)$
17	0	8	$(9^9, 8^9)$
13	2	4, 6	$(8^6, 7^6)$
13	2	0, 11	$(8^9, 6^3)$

The profiles agree within each twin pair. At $p = 13$ the four curves of trace 2 split into two profiles, but that split tracks the group structure — $\mathbb{Z}/12$ for $j = 4, 6$ and $\mathbb{Z}/2 \times \mathbb{Z}/6$ for $j = 0, 11$ — and not the j -invariant, so it is Tier 2 information already accounted for.

3. THE OBSTRUCTION IS A TORSOR COORDINATE, NOT A PAIRING

Remark 3.1 (The proposed explanation is not correct). The barrier was attributed to the Weil pairing being “a non-linear skew-symmetric pairing that cannot be encoded in the abelianized Galois group $\mathbb{I}_K/\overline{K}^\times$ ”. This reasoning does not work. Over a finite field the Galois group acting on $\text{Jac}(C)[n]$ is $\text{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) = \widehat{\mathbb{Z}}$, which is *abelian*; the Weil pairing is Gal-equivariant for that action and is not inaccessible on grounds of non-commutativity. The Tate module with its Frobenius action is precisely the kind of datum an abelian theory can carry — indeed Z_C is its characteristic polynomial.

Theorem 3.2 (The correct obstruction). *Let $\mathcal{E}(\mathcal{O})$ be the set of curves in a fixed ordinary isogeny class with $\text{End} = \mathcal{O}$. Then $\text{Pic}(\mathcal{O})$ acts simply transitively on $\mathcal{E}(\mathcal{O})$, and acts trivially on every invariant of the Galois-module tower $\{E(\mathbb{F}_{q^n})\}_n$, these being $\mathcal{O}/(\phi^n - 1)$ independently of the curve. Consequently the missing datum separating twins is the position of the curve in a $\text{Pic}(\mathcal{O})$ -torsor, and no invariant constructed from the modules — including the Weil pairing, which is likewise $\mathcal{O}$ -determined — can detect it.*

Proof. Simple transitivity is Theorem 1.1(2). Triviality on module invariants is Theorem 1.1(1). A function on $\mathcal{E}(\mathcal{O})$ factoring through the module data is therefore constant, whereas j is injective on $\mathcal{E}(\mathcal{O})$. $\square$

Remark 3.3 (What would be needed). By Theorem 3.2 the separation of twins is equivalent to locating a point in a $\text{Pic}(\mathcal{O})$ -torsor, and the classical object that does this is the j -invariant, i.e. the Hilbert class field of $\mathcal{O}$ and its Galois action. That is still abelian class field theory — but for

the CM field $\mathbb{Q}(\phi)$, not for the function field $K = \mathbb{F}_q(C)$. The two abelian theories are different, and the Bost–Connes system of C carries the second, not the first. This is a sharper diagnosis than “the barrier is non-abelian”, and it suggests where a refinement would have to come from: a system built over the CM order rather than over the curve.

4. THE RESOLUTION BOUNDARY

Theorem 4.1 (Master statement). *For the localized Bost–Connes system of a curve over $\mathbb{F}_q$:*

- *the partition function determines exactly the isogeny class of $\text{Jac}(C)$ (Theorem 4.1 of Paper XVIII);*
- *the Cartan pair determines, through the deep stratum, the compactified Picard group with the degree-labelled point classes $P \mapsto [P] \in \text{Pic}(C)$, hence $\text{Jac}(C)(\mathbb{F}_q)$ and the rational effective cone, strictly refining the previous line (Theorem 3.1 of Paper XIX);*
- *neither determines the curve: by Theorems 1.1 and 2.1, whenever $h(\mathcal{O}) > 1$ there are $h(\mathcal{O})$ curves agreeing in all of the above and differing in j .*

The resolution boundary of the abelian theory is therefore the $\mathcal{O}$ -module tower $\{E(\mathbb{F}_{q^n})\}$, and the residual ambiguity is a $\text{Pic}(\mathcal{O})$ -torsor of size $h(\mathcal{O})$.

5. RETROSPECTIVE

Question	Answer	Reference
do twins exist?	yes, abundantly	Thm. 1.1, Ex. 1.3
is genus two needed?	no: genus one suffices	Rem. 1.2
are twins Tier-1 indistinguishable?	yes	Thm. 2.1
Tier 2 sees only $\text{Jac}(\mathbb{F}_q)$?	no: also $P \mapsto [P]$	Rem. 2.2
are twins Tier-2 indistinguishable?	yes, proved and checked	Thm. 2.1, Rem. 2.3
is the Weil pairing the obstruction?	no; $\text{Gal}(\mathbb{F}_q)$ is abelian	Rem. 3.1
what is the obstruction?	a $\text{Pic}(\mathcal{O})$-torsor coordinate	Thm. 3.2
resolution boundary	$\{E(\mathbb{F}_{q^n})\}$ as $\mathcal{O}$ -modules	Thm. 4.1

The twenty papers have converged on a single principle, and this one states its sharpest form. The arithmetic that a Bost–Connes system carries is the arithmetic of its equilibrium states, and that arithmetic is exactly the Galois-module data of the base: the isogeny class at the thermodynamic level, the Picard group at the level of the Cartan pair. What lies beyond is not a deeper analytic invariant but a torsor, and a torsor has no invariants at all — its points are distinguished only by a coordinate that must be supplied from outside.

That is the honest form of the negative results accumulated here. The K -theory did not fail for lack of computation (Remark 4.5 of Paper II); the von Neumann algebra did not fail for lack of rigidity technique (Theorem 2.1 of Paper IX); the quantum transport inequalities did not fail for lack of a better Dirichlet form (Theorem 4.1 of Paper XVI). In each case the invariant was blind to something that is not an invariant: a measure class, a tail event, and now a torsor coordinate. The one place where the pattern broke was Theorem 4.1 of Paper XVII, where commensurability of the log-norms let the factor retain the temperature — and that too was arithmetic, not technique.

Three questions we would still pursue. Is the analogue of Theorem 3.2 true in higher genus, with $\text{Pic}(\mathcal{O})$ replaced by the relevant class group of the CM order, so that the resolution boundary is uniformly a torsor? Second, Remark 3.3 suggests building a Bost–Connes system over the CM order $\mathcal{O}$ rather than over C , whose symmetry group would be $\text{Pic}(\mathcal{O})$ acting on the twins; does that system exist and does it separate them? Third, the twins of Theorem 1.1 are ordinary; the supersingular case has endomorphisms in a quaternion order, which is where Paper X located the

only viable non-commutative model, and whether the two occurrences of quaternion orders in this programme are related we do not know.

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